

Gourav Barnwal Computer Vision / Machine Learning Engineer

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github.com/GouravBarnwal gouravs-portfolio.vercel.app

PROFESSIONAL SUMMARY

Computer Vision Engineer seeking full-time CV/ML roles, with hands-on experience in feature matching, image retrieval, and anomaly detection — including production systems achieving **0.95+ macro AUROC** and research benchmarking across 9 state-of-the-art vision algorithms under rotation, scale, and low-texture conditions.

PROFESSIONAL EXPERIENCE

- DRDO Young Scientist Laboratory (DYSL-AI), AI Research (Computer Vision) Intern**
• Built a feature-matching and image-retrieval benchmarking pipeline from scratch on TUM RGB-D (613 frames), evaluating **nine algorithms** (SIFT, ORB, SuperPoint+SuperGlue, DISK+LightGlue, ALIKED, XFeat, LoFTR, DeDoDe, RoMa) across rotation, scale, and low-texture robustness conditions
• Conducted DINOv2 CLS-token cosine similarity analysis and identified **LoFTR** as the most robust and fastest performer under degraded conditions, adopted for downstream use
• Evaluated failure modes of text-to-video and vision-language models (Sora, Gemini, and others) across negation, numerical/counting, spatio-temporal, and causal reasoning prompts; extended testing to audio understanding using Whisper transcriptions to probe whether models correctly perceived and reasoned about audio content
- 03/2026 – Present
Bengaluru, India
- PM Accelerator, AI Engineer Intern**
• Built TickerPulse with a **7-member international engineering team**, configuring PostgreSQL and designing RESTful APIs connecting the FastAPI backend to React frontend
- 09/2025 – 12/2025
Boston, Massachusetts, USA (Remote)
- National Institute of Advanced Manufacturing Technology (NIAMT), Ranchi, Software Development Intern**
• Developed a full-stack attendance system using React, Flask, and Firebase serving 150+ students
• Built authentication, attendance workflows, REST APIs, notifications, and SBI Collect payment integration, reducing manual attendance tracking effort for **150+ students**
- 06/2025 – 07/2025
Ranchi, India

PROJECTS

- VISYN — Deep Visual Intelligence for Quality Inspection** , (Python · PyTorch · CV · Anomaly Detection · FastAPI · Docker)
• Built and deployed a production visual anomaly detection system achieving **0.9511 macro AUROC** and 0.9525 macro AP across 6 industrial categories, using MobileNetV3-Small L4/L8 features and category-specific normal reference banks
• Engineered a patch-level comparison pipeline (40/48-channel, 14x14 feature maps → 196 L2-normalized patches per layer), scoring anomalies via TOP-5 patch aggregation, 50:50 L4/L8 fusion, and category-specific P99 thresholds for PASS/REVIEW/FAIL decisions
• Built a deterministic, seed-controlled evaluation framework with per-category and macro AUROC/AP benchmarking, plus patch-based anomaly localization using P95 thresholding and connected-component analysis; productionized with FastAPI/Docker and lazy-loaded reference banks, validated through **14/14 artifact checks** and live inference on Vercel + Render
- DevLens — AI Debugging Tool with Vision Pipeline** , (Python, OpenCV, Tesseract, FastAPI, Next.js, GeminiAPI, DistilBERT, E2B)
• Built an end-to-end pipeline from phone-captured errors through OCR, LLM diagnosis, code localization, fix generation, and sandboxed verification of the generated fix
• Fine-tuned a DistilBERT classifier on 2,069 examples across 22 error types (**82% held-out accuracy**), after identifying and correcting overfitting from an earlier templated dataset
• Designed a 5-candidate OCR ensemble (raw, denoised, binarized, deskewed, deblurred) across up to 4 Tesseract PSM modes, auto-scored to pick the most accurate extraction from noisy phone photos
• Implemented exact/fuzzy/semantic-search code matching and resolved 6+ deployment issues (CORS, cross-platform paths, file watching, cloud provisioning, memory limits) across a live Vercel + Render + Docker + E2B deployment
- FireWatch India — Real-Time Fire Detection & Classification** , (Python · Scikit-learn · Streamlit · Machine Learning)
• Engineered an end-to-end wildfire classification pipeline using 23,000+ MODIS observations, including preprocessing, feature engineering, SMOTE balancing, and model evaluation
• Compared Logistic Regression, Decision Tree, KNN, and Random Forest, achieving **97.8% accuracy** through feature optimization and hyperparameter tuning
• Developed and deployed a Streamlit application with 72-hour live MODIS fire tracking, real-time wildfire prediction, and geospatial visualization

SKILLS

- Computer Vision**
OpenCV, PyTorch, Feature Matching, Image Retrieval, Image Processing, Anomaly Detection, Object Detection, Image Classification, OCR, SIFT, ORB, LoFTR, XFeat, DINOv2
- Backend & Deployment**
FastAPI, Flask, REST APIs, Docker, Git, Linux
- ML/DL**
Scikit-learn, NumPy, Pandas, CNNs, Transfer Learning, Feature Extraction, Model Evaluation
- Frontend & Database**
React, Next.js, Tailwind CSS, PostgreSQL, MongoDB, Firebase
- Programming**
Python, C++, Java, JavaScript, TypeScript

EDUCATION

Bachelor's of Technology in Computer Science & Engineering, Government Engineering College, Ramgarh
CGPA : 7.39/10
12/2022 – 06/2026
Ranchi, India